



977

papers cite one 2024 AI-tutoring paper.

Almost none of them existed when it was published.

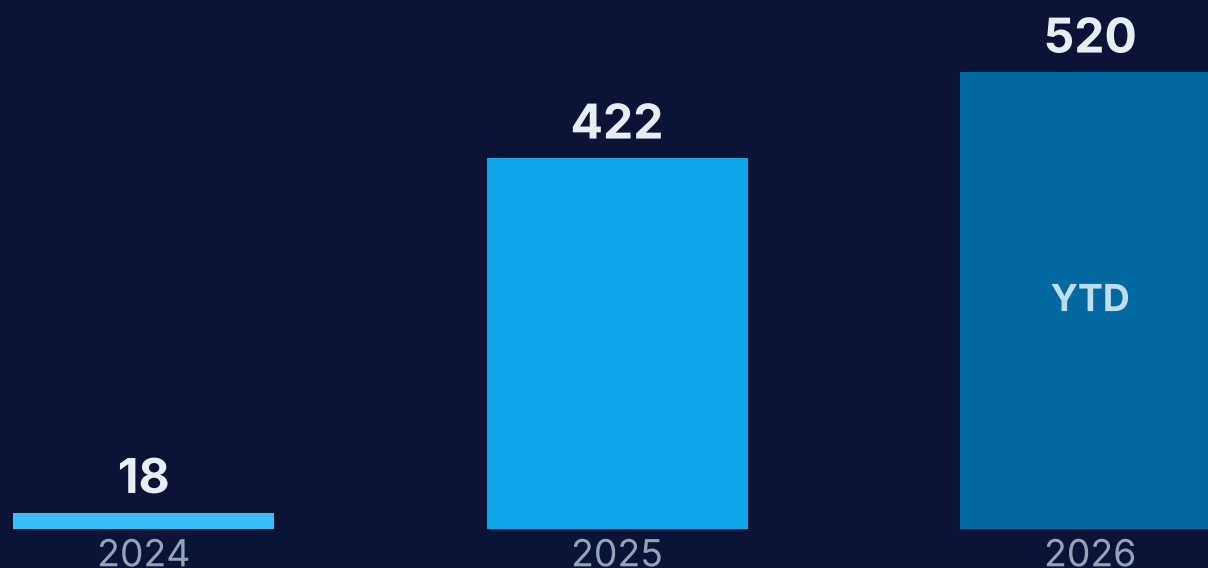
swipe →

96%

*A forward citation tree of a recent paper
is a real-time field-velocity sensor.*

of these papers did not exist
when the seed paper was published.

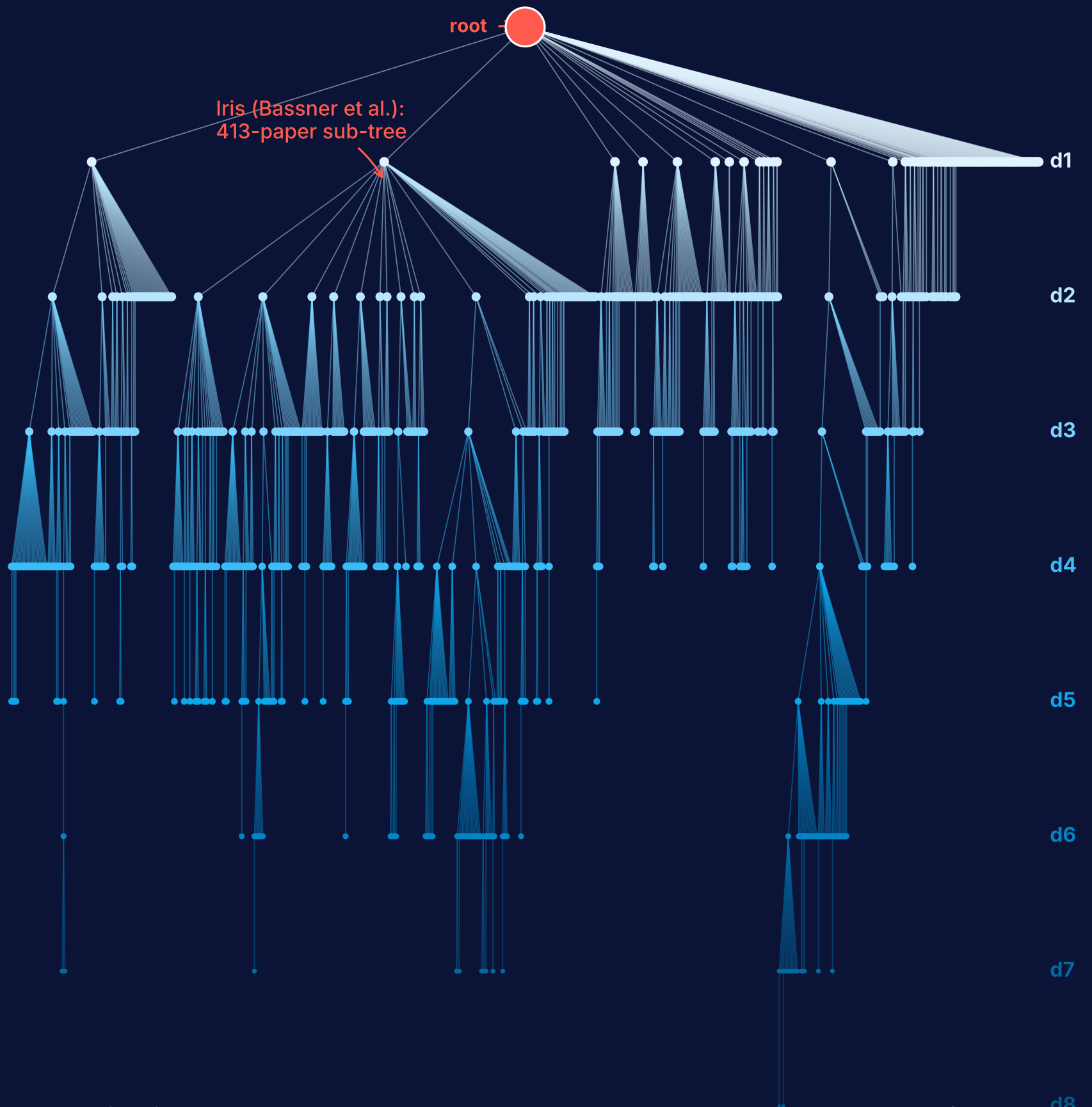
Publication year of all 977 papers in the tree



Crawled June 2026 · 17 papers without a resolved publication year · 2026 bar is year-to-date

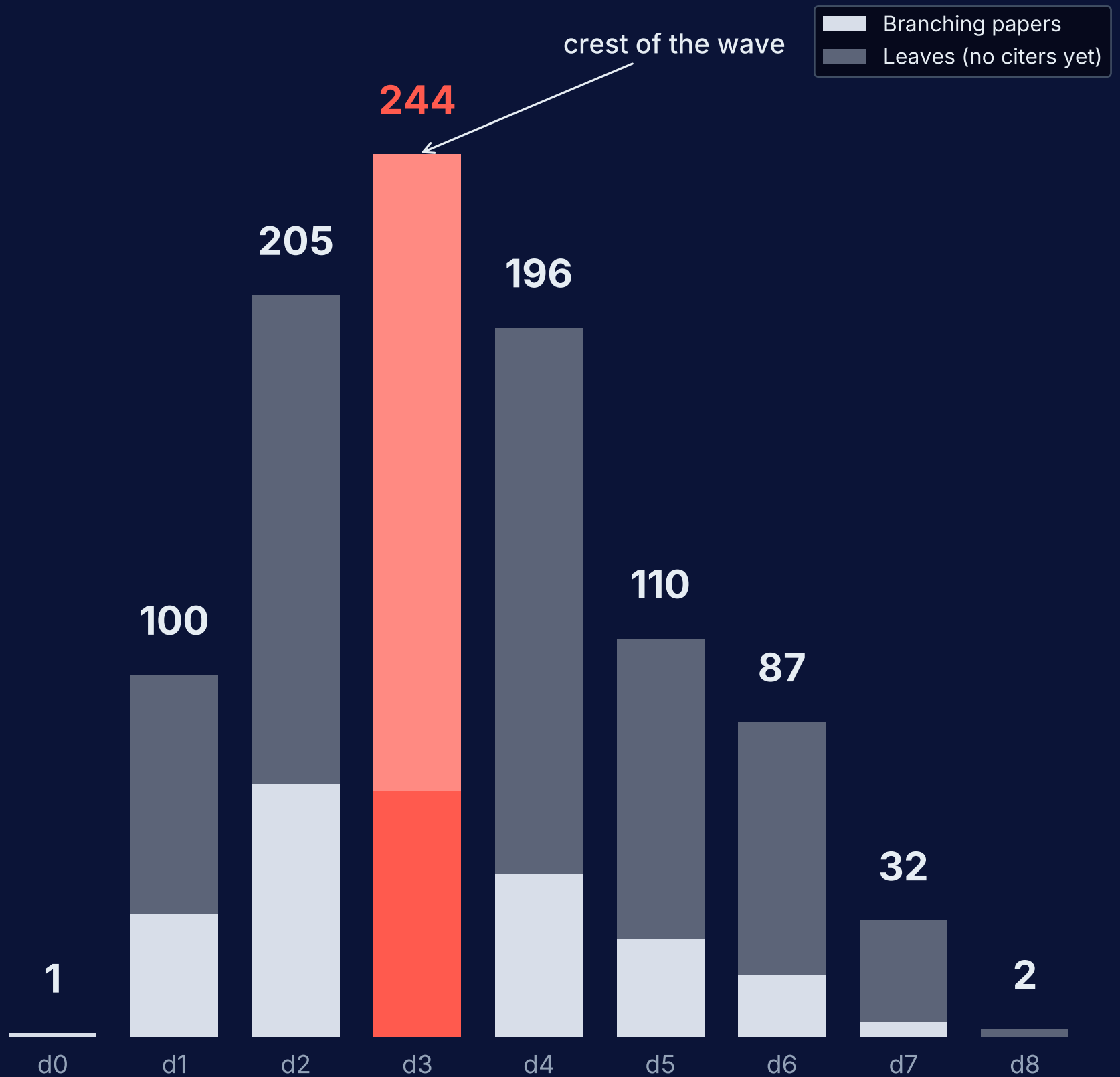
From 1 paper to 977.

But most of the mass lives in one specific layer →



The wavefront

Papers per generation. Depth 3 is where the wave breaks.



Those depth-3 papers are spawning the next wave right now.

@eduardfrankford

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977 papers · 711 leaves · 1062 citation edges

Leaves = no descendants in our crawl (yet).

Where the citations concentrated.

5 papers (of 977) account for 27% of the tree's citation mass.

#1

125

citations

Large language models in computer science education: A systematic literature review

Raihan et al. · 2025

#2

103

citations

Iris: An AI-driven virtual tutor for computer science education

Bassner et al. · 2024

#3

49

citations

Software Engineering by and for Humans in an AI Era

Abrahão et al. · 2025

#4

45

citations

Agentic AI in education: State of the art and future directions

Kostopoulos et al. · 2025

#5

41

citations

Risk management strategy for generative AI in computing education: how to handle the strengths,...

Humble · 2024

977 papers. 96% published after the seed.

Explore every paper in the tree.

↓ Tappable link in the first comment

eduardfrankford.github.io/ai-tutoring-citation-tree



→ Zoom, pan, click any paper for details

→ Search across all 977 titles and authors

SEED PAPER

"AI-Tutoring in Software Engineering Education"

Frankford, Sauerwein, Bassner, Krusche, Breu

ICSE-SEET 2024 · doi.org/10.1145/3639474.3640061

Source: github.com/eduardfrankford/ai-tutoring-citation-tree